

MATHEMATICAL GAMES

A selection of elementary word and number problems

by Martin Gardner

This month I present one of my occasional collections of short problems. As usual, only elementary mathematical skills are called for, and the problems are purposely varied in difficulty and in the type of mathematics involved. The first problem, as a matter of fact, is not mathematical but linguistic; whenever word puzzles have appeared in the department, readers have seemed to enjoy them. The answers to the problems will be given next month.

1.

Dmitri A. Borgmann, the distinguished expert on wordplay, has written a book called *Language on Vacation* (just published by Scribners) that can best be described by borrowing an adjective from the movie *Mary Poppins*—supercalifragilisticexpialidocious. It is the best, most comprehensive book ever written on what Borgmann likes to call “recreational linguistics.” Borgmann’s first objective is “to elevate recreational linguistics to the same high level of esteem now enjoyed by recreational mathematics.” Readers with appetites “jaded by the ubiquitous crossword puzzle,” as Borgmann puts it, will find the book an almost bottomless mine of problems involving palindromes, reversals, anagrams, antigrams, pangrams, geometric word forms, curious letter patterns, numerological aspects of words and hundreds of other oddities previously known only to a small circle of logophiles.

To honor the appearance of this remarkable compendium, here are some unusual word problems that arise in connection with a suggestion made last year by Nicholas Temperley, a student at the University of Cambridge. Temperley proposed that devotees of wordplay produce, as a working tool, an anagram dictionary. Every word in En-

glish is first converted to its “alphabetical anagram,” in which its letters appear in alphabetical order. SCIENTIFIC, for example, becomes CCEFIINST. These alphabetical anagrams are then arranged in alphabetical sequence to form the dictionary. Each entry in the volume is an alphabetical anagram, and under it are listed all the English words that can be formed with those letters. Thus BDEMOOR will be followed by BEDROOM, BOREDOM and all other words formed by those letters. AEIMNNOST will be followed by MINNESOTA and NOMINATES. Some entries even have mathematical interest. AEGILNRT, for instance, will be followed by such mathematical terms as INTEGRAL, RELATING and TRIANGLE as well as other words such as ALTERING. EHNNSTXY will be followed by both NINETY-SIX and SIXTY-NINE. AGHILMORT will have among its anagrams both ALGORITHM and LOGARITHM. If a cross-word puzzle gives the clue BEAN SOUP and an indication that this is an anagram, someone armed with the anagram dictionary need only alphabetize its letters and look up ABENOPSU to find SUBPOENA. If the clue is THE CLASSROOM, it takes only a moment to discover SCHOOLMASTER.

Most of the entries will begin with letters near the front of the alphabet. Temperley estimates that more than half will start with A, which is to say that more than half of all English words contain A. (This is not true of common words, but rarer words tend to be longer and are more likely to have an A.) After I the number of entries will drop sharply. Beyond O the list is extremely short.

Can the reader answer these questions?

1. What will be the dictionary’s last entry? (Place names, such as Uz, the home of Job, are not included.)

2. What will be the first and second entries?

3. What will be the last entry starting with A?

4. What will be the first entry starting with B?

5. An entry, ABCDEFLO, begins with

the first six letters of the alphabet. What is the word?

6. What will be the longest entry that is itself a word? (Short examples include ADDER, AGLOW, BEEFY, BEST, DIPS, FORT.)

7. What will be the longest entry that does not repeat any letter?

2.

A young lady was vacationing on Circle Lake, a large artificial body of water named for its precisely circular shape. To escape from a man who was pursuing her, she got into a rowboat and rowed to the center of the lake, where a raft was anchored. The man decided to wait it out on shore. He knew she would have to come ashore eventually; since he could run four times faster than she could row, he assumed that it would be a simple matter to catch her as soon as her boat touched the lake’s edge.

But the girl—a mathematics major at Radcliffe—gave some thought to her predicament. She knew that on foot she could outrun the man; it was only necessary to devise a rowing strategy that would get her to a point on shore before *he* could get there. She soon hit on a simple plan, and her applied mathematics applied successfully.

What was the girl’s strategy? (For puzzle purposes it is assumed that she knows at all times her exact position on the lake.)

3.

This frustrating topological puzzle calls for a door key and a piece of heavy cord at least a few yards long. Double the cord, push the loop through the keyhole of a door as shown in the top drawing in the illustration on the opposite page, then put both ends of the cord through the projecting loop as in the middle drawing. Now separate the ends, one to the left, the other to the right [*bottom drawing*]. Thread the key on the left cord and slide it near the door, and secure the cord’s ends by tying them to something—the backs of two chairs, for example. Allow plenty of slack.

The problem is to manipulate the key and rope so that the key is moved from spot *P* on the left cord to spot *Q* on the right cord. After the transfer the rope must be looped through the door in exactly the same way as before.

4.

This harmless-looking little problem

in combinatorial geometry, taken from an old, forgotten puzzle book, has more to it than first meets the eye. Forty toothpicks are arranged as shown at the top of the next page to form the skeleton of an order-four checkerboard. The problem is to remove the smallest number of toothpicks that will break the perimeter of every square. "Every square" means not just the 16 small ones but also the nine order-two squares, the four order-three squares and the one large order-four square that is the outside border—30 squares in all.

(On any square checkerboard with n^2 cells the total number of different rectangles is

$$\frac{(n^2 + n)^2}{4},$$

of which

$$\frac{n(n+1)(2n+1)}{6}$$

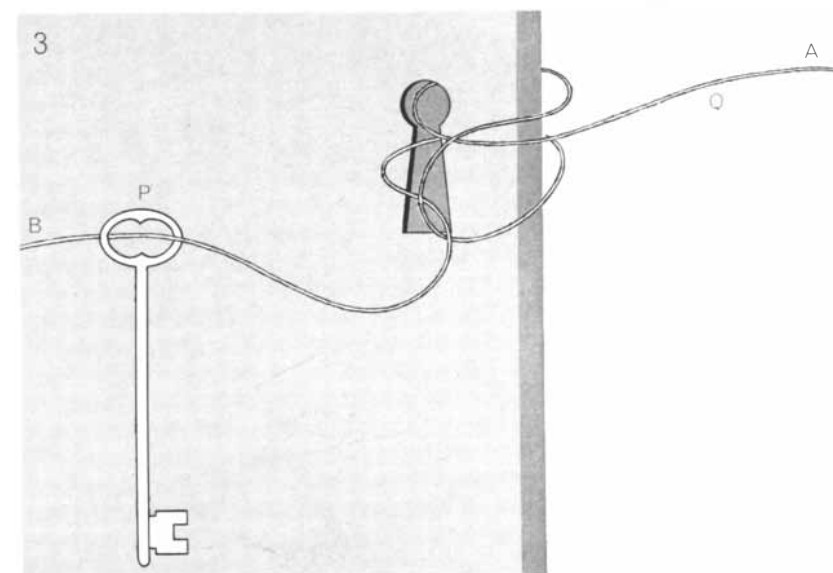
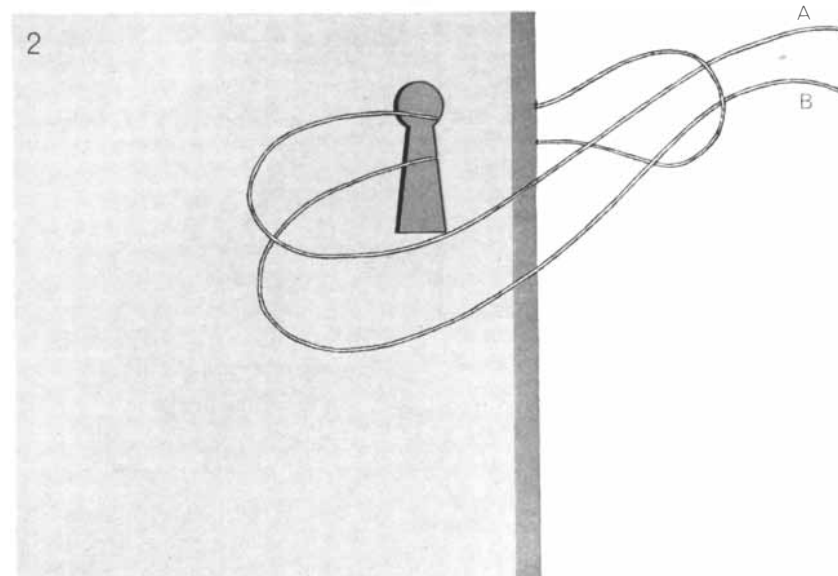
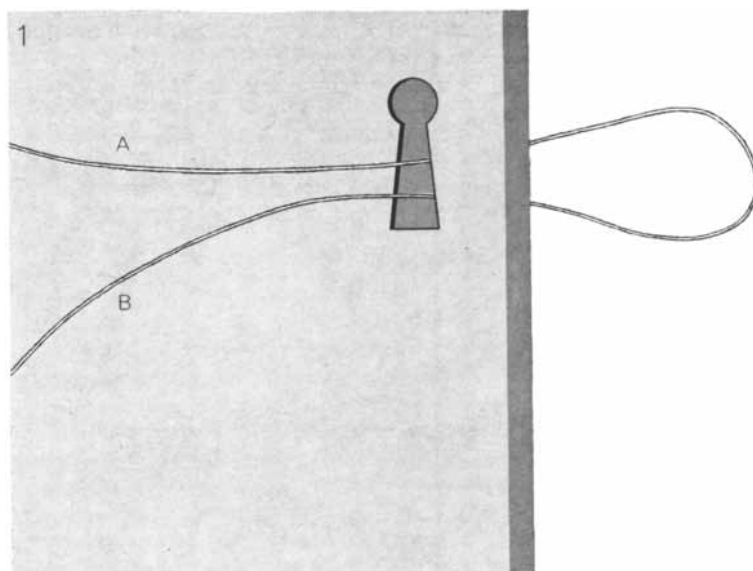
are squares. "It is curious and interesting," wrote Henry Ernest Dudeney, the noted British puzzle expert, "that the total number of rectangles is always the square of the triangular number whose side is n ."

The answer given in the old book was correct, and the reader should have little difficulty finding it. But can he go a step further and state a simple proof that the answer is indeed minimum?

This far from exhausts the puzzle's depth. The obvious next step is to investigate square boards of other sizes. The order-one case is trivial. It is easy to show that three toothpicks must be removed from the order-two board to destroy all squares, and six from the order-three. The order-four situation is difficult enough to be interesting; beyond that the difficulty seems to increase rapidly.

The combinatorial mathematician is not likely to be content until he has a formula that gives the minimum number of toothpicks as a function of the board's order and also a method for producing at least one solution for any given order. The problem can then be extended to rectangular boards and to the removal of a minimum number of unit lines to kill all *rectangles*, including the squares. I know of no work that has been done on any of these questions.

The reader is invited to try his skill on squares with sides from four through eight; I shall present the best available solutions next month. A minimal solution for order-eight, the standard checkerboard (it has 204 different squares), is



Move the key from P to Q



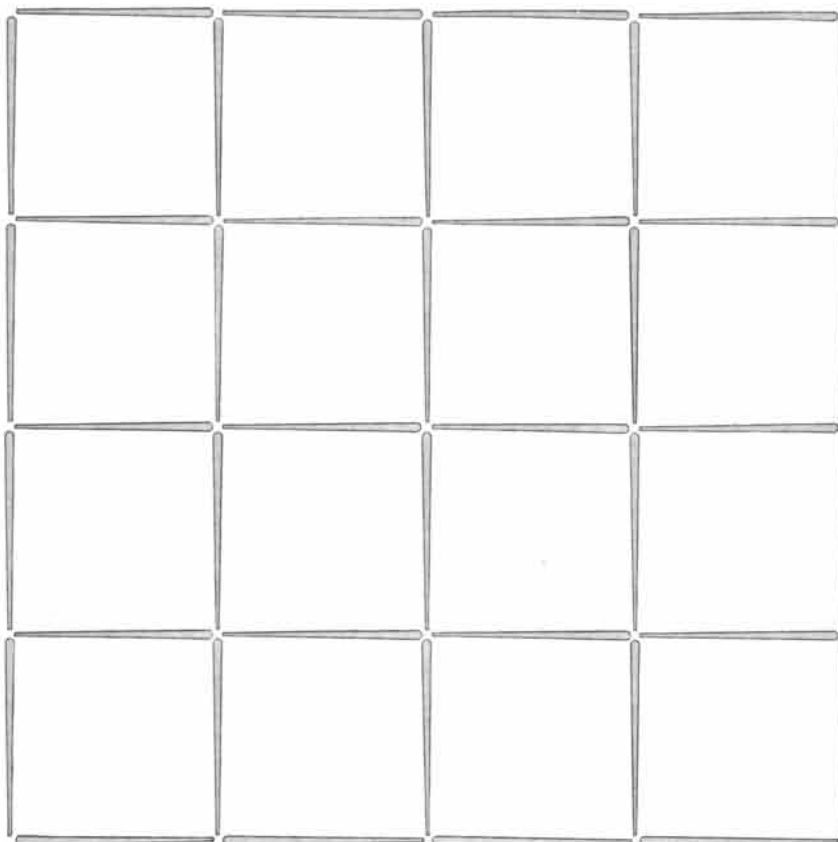
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A toothpick puzzle

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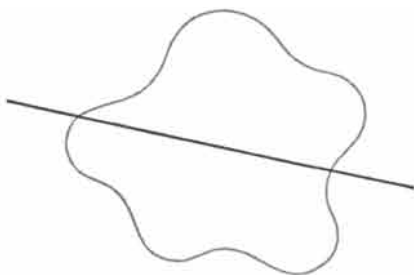
not easy to find, and it will be disclosed next month for the first time.

5.

An infinity of points lie inside the closed curve shown below. Assume that a million of those points are selected at random. Will it always be possible to place a straight line on the plane so that it cuts across the curve, misses every point in the set of a million and divides the set exactly in half so that 500,000 points lie on each side of the line? The answer is yes; prove it.

6.

Gerald L. Kaufman, a New York City



Million-point puzzle

architect and the author of several puzzle books, devised this clever logic problem.

Three playing cards, removed from an ordinary bridge deck, lie face down in a horizontal row. To the right of a King there's a Queen or two. To the left of a Queen there's a Queen or two. To the left of a Heart there's a Spade or two. To the right of a Spade there's a Spade or two.

Name the three cards.

7.

Five paper rectangles (one with a corner torn off) and six paper disks have been tossed on a table. They fall as shown in the illustration on page 120. Each corner of a rectangle and each spot where edges are seen to intersect marks a point. The problem is to find three sets of four "cocircular" points: four points that can be shown to lie on a circle.

For example, the corners of the isolated rectangle [bottom right in illustration] constitute such a set, because the corners of any rectangle obviously lie on a circle. What are the other two sets? This problem and the next are recent inventions of Stephen Barr, au-



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$$Q = \sum g_1 \exp \left[-\frac{E_1}{RT} \right]$$

$$U = \frac{RT^2}{Q} \left[\frac{\partial Q}{\partial T} \right]_V$$

$$\frac{p_2 - p_1}{p_1} = \frac{2 \gamma}{\gamma + 1} (M^2 \sin^2 \beta - 1)$$

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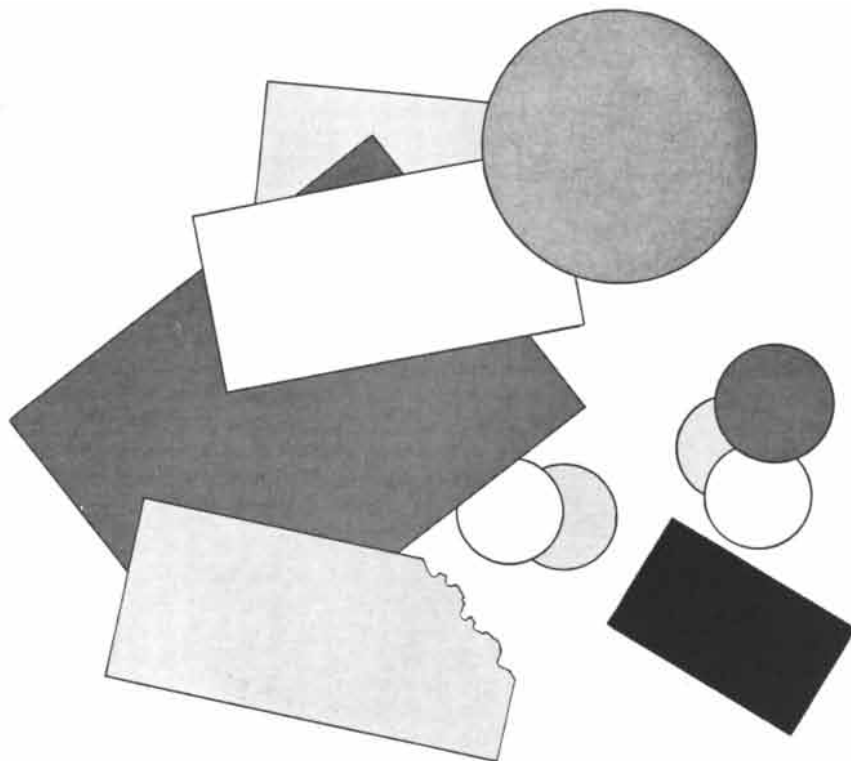
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Find three sets of four cocircular points

thor of *Experiments in Topology* and *A Miscellany of Puzzles: Mathematical and Otherwise*, both published by Crowell. They will appear in a new puzzle book on which he is working.

glasses? (It is assumed that any group of glasses can be tested simultaneously by taking a small sample of liquid from each, mixing the samples and making a single test of the mixture.)

8.

"Mathematicians are curious birds," the police commissioner said to his wife. "You see, we had all those partly filled glasses lined up in rows on a table in the hotel kitchen. Only one contained poison, and we wanted to know which one before searching that glass for fingerprints. Our laboratory could test the liquid in each glass, but the tests take time and money, so we wanted to make as few of them as possible. We phoned the university and they sent over a mathematics professor to help us. He counted the glasses, smiled and said:

"'Pick any glass you want, Commissioner. We'll test it first.'

"'But won't that waste a test?' I asked.

"'No,' he said, 'it's part of the best procedure. We can test one glass first. It doesn't matter which one.'"

"How many glasses were there to start with?" the commissioner's wife asked.

"I don't remember. Somewhere between 100 and 200."

What was the exact number of

9.

Recreational mathematicians have devoted much attention in the past to chessboard "tours" in which a chess piece is moved over the board to visit each square once and only once, in compliance with various constraints. Last year John Harris of Santa Barbara, Calif., devised a fascinating new kind of tour—the "cube-rolling tour"—that opens up a wealth of possibilities.

To work on two of Harris' best problems, obtain a small wooden cube from a set of children's blocks or make one of cardboard. Its sides should be about the same size as the squares of your chessboard or checkerboard. Paint one side red. The cube is moved from one square to an adjacent one by being tipped over an edge, the edge resting on the line dividing the two cells. During each move, therefore, the cube makes one quarter-turn in a north, south, east or west direction.

Problem 1. Place the cube on the northwest corner of the board, red side up. Tour the board, resting once only on every cell and ending with the cube

RECENT findings

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FERROELECTRICS

Ford scientists change certain properties of metals by applying polarized ferroelectric materials

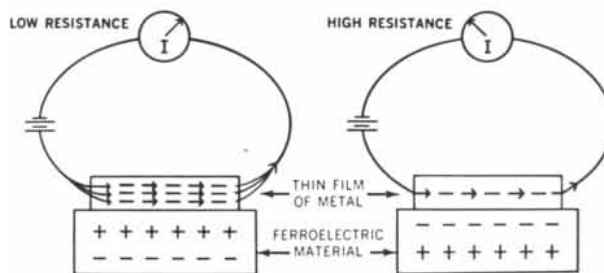
It is well known that many of the physical properties of solids depend upon the number of free charge carriers (usually electrons) per unit volume. This number, and hence the properties, can be changed by placing the solid in an electric field so that more free charges are attracted to or repelled from a given unit volume. Previous attempts to accomplish control over the properties have used direct application of the highest attainable electric fields (approximately 10^7 volts/cm), and the observable effects have always been very small.

Ford Motor Company scientists have developed a method of changing the number of charge carriers in a solid by wedding a thin film of the solid to a crystal of polarized ferroelectric material (barium titanate, triglycine sulfate, and potassium dihydrogen phosphate, for example). Common ferroelectrics, in

a polarized state, carry opposite electrical charges on opposite surfaces of the material, and the surface charge densities correspond to electric fields of 10^9 volts/cm. When a solid connected to a charging source is brought in contact with the charged surface of a polarized ferroelectric, the number of charge carriers per unit volume in the solid will change to cancel the surface charge of the ferroelectric (see diagram). The properties of the solid which depend upon charge carrier density can thus be modified.

This new method of changing the density of charge carriers in a solid is about 100 times more effective and much safer than the method utilizing the direct application of a high voltage electric field. In addition, the studies point the way to several new and useful devices such as computer memory elements, optical display devices and sensitive amplifiers.

Shown here is a diagram of how the resistivity of a thin metal film deposited on a ferroelectric material is modified by the polarization of the ferroelectric substrate. The ferroelectric is shown in a polarized state, i.e. with all its atomic cells or electric dipoles aligned parallel and opposite surfaces carrying opposite surface charges. (Since ferroelectrics are insulators, there is no way for these surface charges to flow together and cancel.) In the left-hand drawing, the charge on the surface adjoining the metal film is positive, thus attracting more negative free charges into the conductor and effectively decreasing its resistance. In the right-hand drawing, the polarization of the ferroelectric has been changed and the surface adjoining the metal film is negatively charged. Negative free charges thus are repelled from the conductor and its resistance is effectively increased.

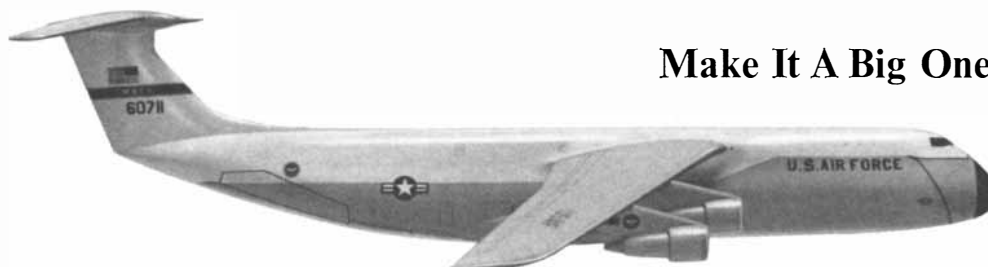


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red side up in the northeast corner. At no time *during* the tour, however, is the cube allowed to rest with the red side up. (NOTE: It is not possible to make such a tour from corner to diagonally opposite corner.)

Problem 2. Place the cube on any cell, an uncolored side up. Make a "re-entrant tour" of the board (one that visits every cell once and returns the cube to its starting square) in such a way that at no time during the tour, including at the finish, will the cube's red side be up.

Both problems have unique solutions, not counting rotations and reflections of the path.

The first of last month's pentomino problems was to pick, from among the 33 different two-move pentomino games on the four-by-four board, the game that has only one winning reply by the second player. It is the game that was numbered 26 and is shown again at A in the illustration below. The first play leaves a space on the right that can be filled only by the L-pentomino. But if the L is placed in that space, the first player can win by playing on the left. If, on the other hand, the second player puts any piece except the L on the left, the first player can win by playing the L on the right. To win,

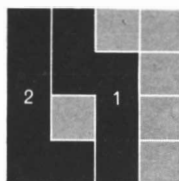
therefore, the second player must place the L on the left as shown.

On the five-by-five board the first player has an obvious win by playing the I-pentomino in the center, as shown at B below. His opponent must play on one side and the first player then wins by playing on the other side. The shortest possible game on this board has two moves [C] and the longest has five [D]. The short-game pattern is unique, but there are many solutions to the long game.

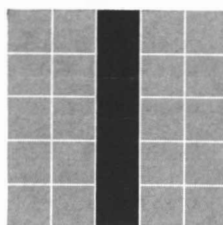
The first player can win on the five-by-six board only if his first move is the one shown at E. There is no simple proof, and space does not allow showing correct responses to all possible second moves.

The Y-pentomino is rectifiable. The drawing at F shows the smallest rectangle that can be formed with replicas of this piece. The pattern illustrated is one of four possible rectangles.

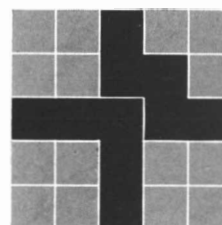
I hope readers will understand that it was not possible to reply to some 100 correspondents who sent correct solutions to the interplanetary-code message in this department for August. More than half of these solutions included a correct interpretation of the final statement, identifying the earth as the source of the message.



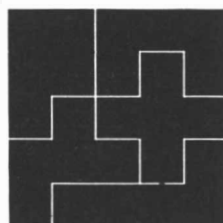
A



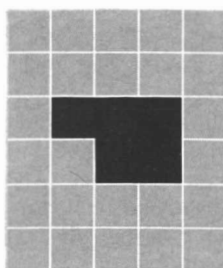
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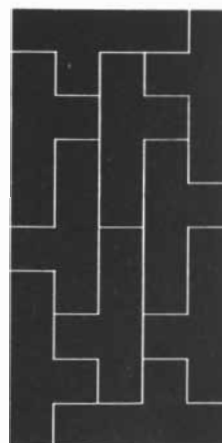
C



D



E



F

Answers to pentomino problems